

## Homework 3

For Monday, September 21, 2026.

Give exact answers and justify your reasoning. Do not use electronic tools. You may use the limit laws and examples covered in Lectures 3 and 4. When a proof directly from the definition is requested, use the epsilon–delta definition. All functions are real-valued. Unless otherwise specified, a function whose limit at  $a$  is considered is defined on an open interval containing  $a$ , except possibly at  $a$  itself. Conditions on  $x$  refer to points in the relevant domain, or common domain when functions are combined.

1. Suppose  $\lim_{x \rightarrow a} f(x) = 2$  and  $\lim_{x \rightarrow a} g(x) = -1$ . Find

$$\lim_{x \rightarrow a} \frac{f(x)^2 + 3g(x)}{2f(x) - g(x)}.$$

Explain why the denominator is nonzero sufficiently near  $a$ , with  $a$  excluded.

2. Find

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}.$$

Explain why the quotient law cannot be applied directly to the original numerator and denominator, and justify your calculation.

3. Find all real numbers  $c, d$  for which

$$\lim_{x \rightarrow 2} \frac{x^2 + cx + d}{x - 2} = 5.$$

Justify that your values work and that no others do. *Hint:* if this quotient has the stated limit, multiply it by  $x - 2$  and use the product law.

4. Suppose

$$\lim_{x \rightarrow a} (2f(x) + g(x)) = 7, \quad \lim_{x \rightarrow a} (f(x) - g(x)) = 2.$$

Prove that  $f$  and  $g$  each have a limit at  $a$ , and find both limits. Do not assume their existence at the start.

5. Prove directly from the epsilon–delta definition that

$$\lim_{x \rightarrow 1} \frac{1}{x + 1} = \frac{1}{2}.$$

Give an explicit choice of  $\delta$  for each  $\varepsilon > 0$ , and show how you keep the denominator away from zero.

6. Suppose  $\lim_{x \rightarrow a} f(x) = -3$ . Prove that there exists  $r > 0$  such that, whenever  $0 < |x - a| < r$ ,

$$-4 < f(x) < -2, \quad \frac{1}{4} < \frac{1}{|f(x)|} < \frac{1}{2}.$$

7. Construct a function  $f : \mathbb{R} \rightarrow \mathbb{R}$  with a finite limit at zero but with no bound on  $|f(x)|$  over all of  $\mathbb{R}$ . Justify both properties; for the second, show that for every  $B > 0$  there is an  $x \in \mathbb{R}$  with  $|f(x)| > B$ .

8. Suppose that whenever  $0 < |x| < 1$ ,

$$|f(x) - 2| \leq 3|x|, \quad |g(x) + 1| \leq 2|x|.$$

Prove directly that  $\lim_{x \rightarrow 0} f(x)g(x) = -2$ , giving an explicit  $\delta$  for every  $\varepsilon > 0$ .

9. Suppose  $|f(x)| \leq 4$  whenever  $0 < |x - a| < r$ , for some  $r > 0$ , and  $\lim_{x \rightarrow a} g(x) = 0$ . Prove directly from the definition that  $\lim_{x \rightarrow a} f(x)g(x) = 0$ , without assuming that  $f$  has a limit. Then use your argument to find

$$\lim_{x \rightarrow 0} x \sin(1/x).$$

10. Suppose  $\lim_{x \rightarrow a} f(x) = L$ . Prove directly from the definition that

$$\lim_{x \rightarrow a} |f(x)| = |L|.$$

You may use the inequality  $||u| - |v|| \leq |u - v|$  for real  $u, v$ .

11. Let  $L$  be any real number. Construct polynomials  $P, Q$  such that

$$\lim_{x \rightarrow 0} P(x) = 0, \quad \lim_{x \rightarrow 0} Q(x) = 0, \quad \lim_{x \rightarrow 0} \frac{P(x)}{Q(x)} = L,$$

with  $Q(x) \neq 0$  for every  $x \neq 0$ . Verify your construction.

12. Define

$$f(x) = \begin{cases} x + 1, & x < 0, \\ 0, & x = 0, \\ 3 - x, & x > 0. \end{cases}$$

Prove that  $f$  has no finite limit at zero. For each candidate  $L$ , use the negated definition to give a fixed  $\varepsilon_0 > 0$  and a suitable  $x$  for every  $\delta > 0$ .

13. Prove that  $\cos(1/x^2)$ , defined for  $x \neq 0$ , has no finite limit as  $x \rightarrow 0$ . Use the negated definition and give explicit points in every punctured interval  $0 < |x| < \delta$ . You may use  $\cos(2\pi n) = 1$  and  $\cos((2n + 1)\pi) = -1$  for positive integers  $n$ .
14. Construct functions  $f, g : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$  such that neither has a finite limit at zero, but both  $f + g$  and  $fg$  do. Find the two limits and justify all the stated properties.

15. Suppose  $f(x) \neq 0$  for every  $x$  sufficiently near  $a$ , with  $a$  excluded, and

$$\lim_{x \rightarrow a} \frac{1}{f(x)} = 0.$$

Prove, using the limit laws, that  $f$  cannot have a finite limit at  $a$ .